

Practical 1 - Introduction to Survival Analysis

Survival Analysis

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Datasets required: `pbcbase`, `surv_data_practical1`, `whitehall`

R packages required: `survival`, `mexhaz`, `flexsurv`

```
library(survival)
library(mexhaz)
library(flexsurv)
```

Introduction

This practical is in three parts.

Part A. Looks at some fundamentals of working with survival data. Using `pbcbase`.

Part B. You will derive the formula for the maximum likelihood estimator of the hazard λ under an exponential distribution, and then estimate λ from some data. Using `surv_data_practical1`.

Part C. You will derive the formula for the maximum likelihood estimator of the hazard λ under an exponential distribution, and then estimate λ from some data. Using `whitehall`.

Aims

The aims of this session are:

- Understand how survival data are stored and set-up for analysis.
- Derive the formula for the maximum likelihood estimator of the hazard rate parameter λ from an exponential distribution.
- Be able to obtain the maximum likelihood estimate $\hat{\lambda}$ from data.

Part A: Using survival data

The first dataset we will use is called `pbcbase`. This is data from a multicentre, double-blind, clinical trial for the treatment of Primary Biliary Cirrhosis (PBC). There are 184 individuals in the data and 17 variables; the variables of interest in this session are described below.

Variable	Description
<code>id</code>	Unique identifier for each participant
<code>datein</code>	Date person entered the study
<code>dateout</code>	Date of the end of follow-up due to either death or censoring
<code>d</code>	Event indicator at the end of follow-up: 0=alive (censored), 1=dead
<code>time</code>	Follow-up time in years

Question 1

Open the `pbcbase` dataset and familiarise yourself with the key variables.

Question 2

The `datein` and `dateout` variables are stored as text in the csv file. Start by formatting these as dates in R as follows: `pbcbase$datein=as.Date(pbc$datein,"%d/%b/%Y")` `pbcbase$dateout=as.Date(pbc$dateout,"%d/%b/%Y")` Take a look at what has happened to `datein` and `dateout`. They are now stored in the default form: YYYY-MM-DD. Behind the scenes in R the dates are now recorded as numbers. The number used by R to encode a particular date is as the number of days after 1st January 1970. So 1st January 1970 takes value 0, 2nd January 1970 takes value 1, etc.

The `lubridate` package is also useful for handling dates, but we do not use it in this module. Here is a helpful blog on dates and times in R: <https://www.gormananalysis.com/blog/dates-and-times-in-r-without-losing-your-sanity/>

Question 3

- How many people died in the dataset, and how many were censored?
- What was the earliest date of entry to the study? And the latest?
- What was the earliest date of exit from the study? And the latest?
- What happened to the first two people in the data (`id=5` and `id=23`).

Question 4

The fact that R stores dates as numbers makes calculating the time between two dates simple. To calculate the length of time each person was followed in the study (which we call `days_in_study`) we can type: `pbcs$days_in_study = as.numeric(pbc$dateout - pbc$datein)`

Compare this new variable to the `time` variable to make sure this has done what you expect.

Question 5

What do you think is the appropriate time origin in the PBC study?

In R the `survival` package contains the key tools for analysis of survival data. The `Surv()` function is fundamental to survival analysis in R. It is used in functions such as model formulas to specify the time periods over which people are observed and to specify which individuals have the event and which are censored. Try the following: `Surv(time=pbc$time,event=pbc$d)`

Take a look at the output. What do the `+` symbols represent? This method uses the date of randomisation as the time origin (i.e. time since randomisation is the time scale) and specifies the event indicator. As the `time` variable is in years, this will be the units in which time is used in any analysis in which the above `Surv()` function is included.

Another way of specifying the `Surv()` function is by using the dates:

```
Surv(time=as.numeric(pbc$dateout),event=pbc$d,origin=as.numeric(pbc$datein))
```

In this version, the time of the end of follow-up is provided as the date each person left the study, and the options tell R the time origin (the date of joining the study, in this case), and the event indicator. Because the entry and exit dates (`datein` and `dateout`) are stored in number of days, the unit of analysis will be days. Note that the `Surv()` function does not accept times that are formatted as dates, which is why we use `as.numeric` above.

Part B: Fitting models to survival data

Question 1

Consider a study of n individuals. Let t_i denote the event or censoring time for individual i and let δ_i be the indicator of whether the person died (meaning t_i is their time of death) or was censored (meaning t_i is their censoring time, i.e. they were known to be alive at this time).

Write down the likelihood for survival data (including censoring) assumed to follow an exponential distribution with hazard λ . Derive a formula for the maximum likelihood estimate for λ .

Question 2

We will now use the formula derived for $\hat{\lambda}$ in a simulated dataset called `surv_data_practical1`. There are just two variables: `survtimes` contains the survival time for each participant, and `d` the outcome. In this dataset all participants experienced the event (`d=1`). The data were simulated for 100 individuals, with event times generated from an exponential distribution.

Load the data using

```
dta=read.table("datasets/surv_data_practical1.csv",sep="," ,header=T)
```

Using your formula from Part B, Question 1 calculate the maximum likelihood estimate for λ and interpret the estimate.

Question 3

We will fit an exponential model to these data to confirm the maximum likelihood estimate for λ that we calculated above. Compare the results with what you found above. In R exponential models can be fitted using the `survreg` function from the `survival` package:

```
exp.model = survreg(Surv(survtimes)~1,dist="exponential",data=mydata)
summary(exp.model)
```

It is important to note that the `survreg` function with `dist="exponential"` estimates $-\log \lambda$ rather than λ (See Table 3.1 in the lecture notes).

Another option is the `mexhaz` package, where we can fit an exponential distribution using:

```
exp.model2 = mexhaz(Surv(survtimes)~1,base="pw.cst",data=dta)
exp.model2
```

The `flexsurv` package provides another way of fitting an exponential model, as follows:

```
exp.model3 = flexsurvreg(Surv(survtimes)~1,dist="exponential",data=mydata)
exp.model3
```

Compare the output given by the different functions.

Part C: Whitehall study data

The Whitehall Study is a cohort study, which was started in 1967 and is a cohort of male civil servants employed in various government departments in London. It has been used to study risk factors for mortality in men. We will focus on deaths due to coronary heart disease (CHD). The dataset contains 1677 individuals and 14 variables, with the variables used in this practical summarised below:

Variable	Description
id	Unique identifier for each participant
timebth	Date of birth
timein	Date person entered the study
timeout	Date of the end of follow-up due to either death or censoring
chd	Event indicator at the end of follow-up: 0=alive or died from other cause, 1=death due to coronary heart disease

Question 1

- Load the data and format `timebth`, `timein` and `timeout` as dates.
- What is the latest date of the end the follow-up? How many people had their end of follow-up on this date? What could the explanation for this be?

Question 2

This is an example of data where event times could be left-truncated, meaning that not everyone is observed from the time origin of interest. In this type of study we could be interested in the date of joining the study as the time origin, or the date of birth as the time origin, depending on our question.

Compare the following four ways of using `Surv()` for this data.

```
#(i) time origin: date of joining cohort
#time scale: time in study
Surv(time=as.numeric(whl$timeout),event=whl$chd,origin=as.numeric(whl$timein))

#(ii) time origin: date of birth
#time scale: age
#delayed entry: ignored (not a good idea)
Surv(time=as.numeric(whl$timeout),event=whl$chd,origin=as.numeric(whl$timebth))
```

```
#(iii) time origin: date of birth
#time scale: age
#delayed entry: left truncation accommodated by providing the
# information that people join the study at timein
Surv(time=as.numeric(whl$timein),
      time2=as.numeric(whl$timeout),
      event=whl$chd,
      origin=as.numeric(whl$timebth))
```